

WORK, POWER AND ENERGY

- **WORK**

- When we applying a force on a body if body is displaced then we said that work is done.
i.e, $\text{Work} = \text{Force} \times \text{Displacement towards the force}$
- Work is denoted by 'W'.
- If Force 'F' and displacement 'S' are in the same direction.
Thus, $W = F \times S$
- Work is a scalar quantity because it is scalar product of Force and displacement.
- The S.I unit of work = S.I unit of F $\times$ S.I unit of D
= N m

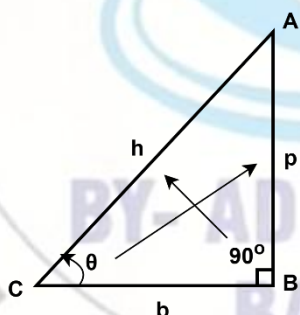
Note:

- 1 Nm = 1 joule

$\therefore$ The S.I unit of work is 'Joule' and it is denoted by 'J'.

- The other unit of work is 'Calorie'.
i.e, 1 Calorie = 4.182 J
1 Cal $\approx$ 4.2 J
- The magnitude of work done can be negative, positive or, zero.

Only for Understand



P = perpendicular
b = base
h = hypotenuse

$$\Rightarrow h^2 = p^2 + b^2$$

$$\therefore \sin \theta = \frac{p}{h},$$

$$\cos \theta = \frac{b}{h}$$

[Trigonometrical function]

Value of Sin θ and Cos θ at particular angle

$$\cos 0^\circ = 1$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\cos 90^\circ = 0$$

$$\cos 180^\circ = -1$$

$$\sin 0^\circ = 0$$

$$\sin 30^\circ = \frac{1}{2}$$

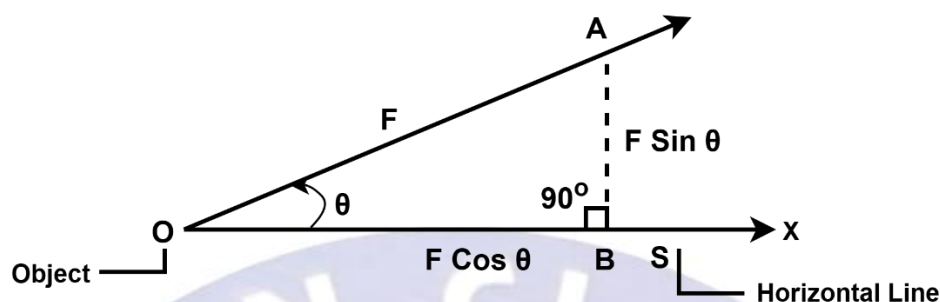
$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\sin 90^\circ = 1$$

$$\sin 180^\circ = 0$$

- **Work done is derived as**
- Let the angle between Force (F) and displacement (S) is θ .



If the force applied on the object is F in the direction of OA and displacement (S) in the direction of OX .

Angle between F and S ($\angle AOX$) = θ

Let Horizontal component of Force (F) is $F \cos \theta$ in the direction of displacement (S).

$$\text{i.e., } \frac{OB}{OA} = \cos \theta$$

$$\Rightarrow OB = OA \times \cos \theta$$

$$\Rightarrow OB = F \cdot \cos \theta$$

Similarly,

The vertical component of Force (F) is $F \cdot \sin \theta$ in the $\perp$ direction of displacement (S).

$$\text{i.e., } \frac{AB}{OA} = \sin \theta$$

$$\Rightarrow AB = OA \cdot \sin \theta$$

$$\Rightarrow AB = F \cdot \sin \theta$$

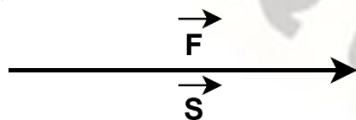
$\therefore W = \text{Force in the direction of displacement (S)} \times \text{Displacement (S)}$

$$\therefore W = F \cos \theta \cdot S$$

$$\Rightarrow \boxed{W = F \cdot S \cos \theta}$$

Case 1: When the angle between F and S is $\theta = 0^\circ$.

i.e., F and S are in the same direction.



$$\therefore W = F \cdot S \cos \theta$$

$$\therefore W = F \cdot S \cos 0^\circ$$

$$= F \cdot S \times 1 \quad [\because \cos 0^\circ = 1]$$

$$\Rightarrow W = F \cdot S \quad [W = F \times S]$$

Note:

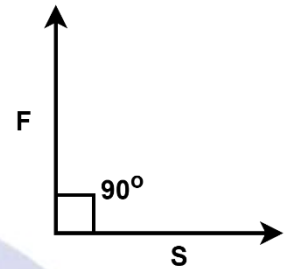
- In this case object must be displaced otherwise work is not done.

For example:-

- ✓ When we pulled / pushed a gate horizontally.
- ✓ When we pulled / pushed a car horizontally.

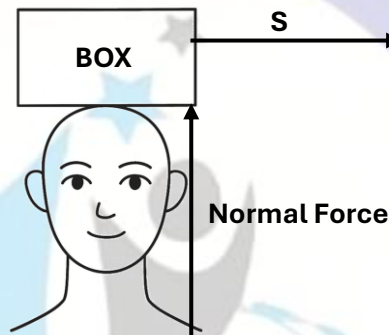
Case 2: When angle between F and S is 90° .

$$\begin{aligned}\therefore W &= F \cdot S \cos\theta \\ \Rightarrow W &= F \cdot S \cos 90^\circ \\ \Rightarrow W &= F \cdot S \times 0 = 0 & [\cos 90^\circ = 0] \\ \Rightarrow W &= 0\end{aligned}$$



For example:-

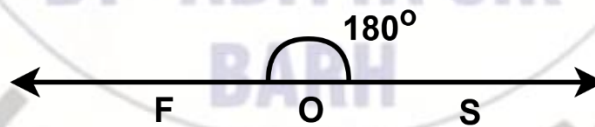
- ✓ In the case of coolie



Note:

- Every object lifts on the head and moves horizontally then work done is 0.

Case 3: When angle between F and S is 180° .

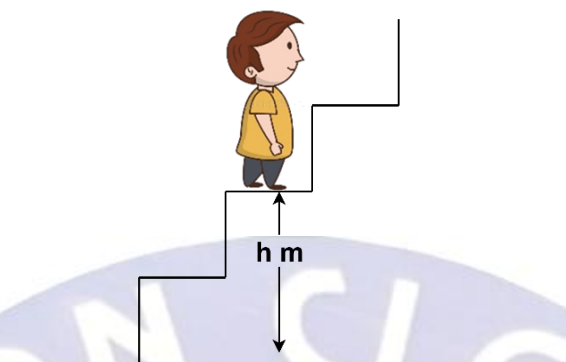


$$\begin{aligned}\therefore W &= F \cdot S \cos\theta \\ \Rightarrow W &= F \cdot S \cos 180^\circ \\ \Rightarrow W &= F \cdot S (\times -1) & [\cos 180^\circ = -1] \\ \Rightarrow W &= -F \cdot S\end{aligned}$$

For example:-

- ✓ In the case of breaking of car, bike, bus etc. and on walking (Friction).

• Work done against the gravity



$$W = mgh$$

Here, the angle between weight (Force) and Height is 0° .

$$\therefore W = F \cdot S \cos\theta$$

$$\Rightarrow W = mgh \cdot \cos 0^\circ \quad [F = mg]$$

$$\Rightarrow W = mgh \times 1 \quad [\cos 0^\circ = 1]$$

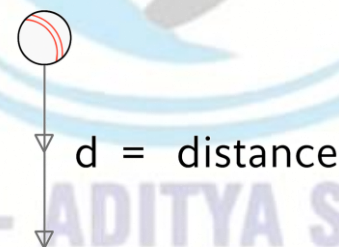
$$\Rightarrow W = mgh$$

For example:-

- ✓ In the case of climbing stairs, lifting a chair, lifting a dumbbell etc.

Note:

1. Work done in moving a ball fall down.

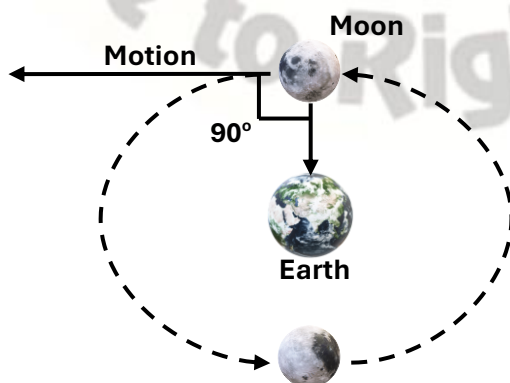


Here,

$$W = F \cdot d$$

$$W = mg \cdot d \quad [F = mg]$$

- 2.



- In the case of planets moves around the sun moon / satellites move around the Earth, the Work done is zero.

For example:-

Q. A person of mass 60 kg lifts on a stair at the height of 5 m. Find the work done by person.

Ans $m = 60 \text{ Kg}$

$$g = 9.8 \text{ m/s}^2$$

$$h = 5 \text{ m}$$

$$\begin{aligned}\therefore W &= m \times g \times h \\ &= 60 \times 9.8 \times 5 \\ &= 300 \times 9.8 \\ &= 2940.0 \\ W &= 2940 \text{ Joule}\end{aligned}$$

Q. A trolley bag is pulling with force of 10 N inclined with Horizontal surface by angle 45° . If trolley is displaced to 100 m. Find the work done.

Ans $F = 10 \text{ N}$

$$S = 100 \text{ m}$$

$$\theta = 45^\circ$$

$$\begin{aligned}\therefore W &= F \cdot S \cdot \cos\theta \\ \Rightarrow W &= 10 \times 100 \times \cos 45^\circ \\ \Rightarrow W &= 1000 \times \frac{1}{\sqrt{2}} \\ \Rightarrow W &= \frac{1000}{\sqrt{2}} \text{ Joule}\end{aligned}$$

ENERGY

- The capacity of work done is called energy.
- Energy is also a scalar quantity.
- The S.I unit of energy is 'Joule'.
- The biggest unit of energy is 'Kilojoule' and it is denoted by 'kJ'.

$$1 \text{ kJ} = 1000 \text{ J}$$

Note:

- Joule name in honored of British physicist "James Prescott Joule".

- Energy is denoted by "E".

Types of energy

1. Kinetic energy.
2. Potential energy.
3. Chemical energy.
4. Heat energy.
5. Light energy.
6. Sound energy.
7. Electrical energy.
8. Nuclear energy.
9. Mechanical energy.
10. Muscular energy.
11. Elastic energy.
12. Solar energy.
13. Wind energy.
14. Tidal energy.

1. Kinetic energy

- The energy of an object due to its velocity / speed is called Kinetic energy.

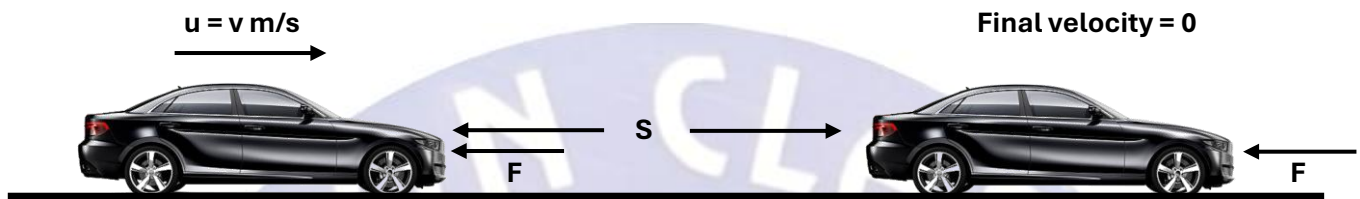
➤ Kinetic energy is denoted by "K.E".

➤ The formula of K.E = $\frac{1}{2}mv^2$

Where, m = mass of body
 V = velocity of body

Proof:

➤ Let a body of mass = m kg and initial velocity = v m/s



➤ If a body (car) applies break, then friction (F) acts opposite to velocity of car.

$$\therefore v^2 = u^2 + 2as$$

$$\Rightarrow 0^2 = v^2 + 2as$$

$$\Rightarrow -2as = v^2$$

$$\Rightarrow a = \frac{-v^2}{2s}$$

Negative acceleration means object is in retardation

$$\therefore \text{The acceleration in the body, } a = \frac{-v^2}{2s}$$

$$\therefore W = F \cdot S \cos\theta$$

Here, work done is in the form of Kinetic energy

$$\therefore K.E = F \cdot S \cos\theta$$

$$[W = K.E]$$

$$\Rightarrow K.E = ma \times S \cos 180^\circ$$

$$[\therefore F = ma]$$

$$\Rightarrow K.E = m \times \frac{-v^2}{2s} \times S \times -1$$

$$[\therefore \cos 180^\circ = -1, a = \frac{-v^2}{2s}]$$

$$\Rightarrow K.E = \frac{1}{2}mv^2$$

Hence, the proved.

For example:-

Q. A cyclist of mass 50 kg moves with 10 m/s. Find the K.E.

Ans $\therefore m = 50$ kg, $v = 10$ m/s

$$\therefore K.E = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 50 \times 10 \times 10 = 25 \times 100$$

$$= 2500 \text{ J}$$

2. Potential energy

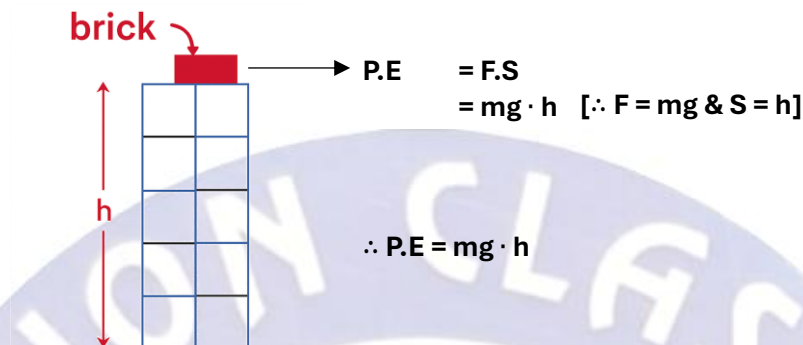
➤ The energy of an object because of its either position or shape is called potential energy.

Types of Potential energy

- a) **Gravitational Potential Energy.**
- b) **Elastic Potential Energy.**

a) **Gravitational Potential Energy**

- The Potential Energy due to height, above the earth surface is called gravitational potential energy.



b) **Elastic Potential Energy**

- The extra energy stored in a stretched / compressed spring / rubber band is called its elastic potential energy.

For example:-

- ✓ In the case of Catapult (Gulel).
- ✓ In the case of Bow and Arrow.

3. **Mechanical Energy**

- The sum of Kinetic Energy and Potential Energy is known as Mechanical Energy.
i.e, Mechanical Energy = K.E + P.E

Note:

1. If one of the either Kinetic Energy or, Potential Energy is zero then Mechanical Energy is also zero.

For example:-

- ✓ Moving ceiling fan, The flying aeroplane, The flying bird, Ridding a bicycle.
2. Work done by external forces on a system is equal to the increase in the system energy.

• **Law of conservation of energy**

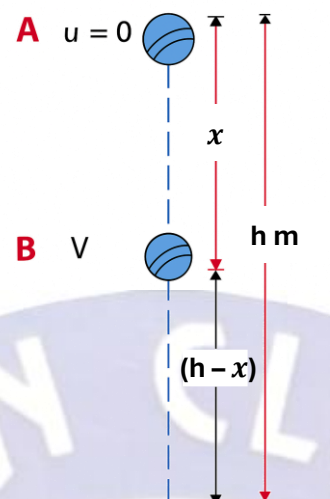
- Energy is neither be created nor be destroyed it can only transform from one form to another form; thus energy is conversed this law is known as law of conversation of energy /principal of conversation of energy.

▪ **A/C to the principal of conservation of energy**

- The total energy in the universe remains constant it can only transform from one body to another body, but it is only applicable for isolated body and system

A proof of Conservation of Energy

Proof: Let a body of mass m kg and rest at a point A. h meter above the ground and B is a point $(h - x)$ meter above the earth.



Case 1: When body at point A

$$\therefore \text{P.E} = mgh \quad \& \quad \text{K.E} = \frac{1}{2}mv^2$$

$$\text{K.E} = \frac{1}{2}m \times 0^2 = 0$$

$\therefore$ Energy on point A (E_A) = P.E + K.E

$$\Rightarrow E_A = \text{P.E} + \text{K.E}$$

$$\Rightarrow E_A = mgh + 0$$

$$\Rightarrow E_A = mgh \quad \dots\dots\dots (i)$$

Case 2: When body at point B

$$\because v^2 = u^2 + 2gs$$

$$\Rightarrow v^2 = 0^2 + 2gx \quad [\because S = x]$$

$$\Rightarrow v^2 = 2gx$$

and $\text{P.E} = mg(h - x)$

$$\& \quad \text{K.E} = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times m \times 2gx$$

$$\& \quad \text{K.E} = mgx$$

$\therefore$ The Energy on point B (E_B)

$$\therefore E_B = \text{P.E} + \text{K.E}$$

$$= mg(h - x) + mgx$$

$$= mgh - mgx + mgx$$

$$E_B = mgh \quad \dots\dots\dots (ii)$$

From equation (i) and (ii)

$$E_A = E_B$$

Hence, the proved.

- **Work, Energy theorem**

- According to work energy theorem, "Work done is sum of all forces on the particles or, body equal to change in Kinetic energy of the particle". This principle is also known as principal of work and kinetic energy.

$$\begin{aligned}\text{i.e, Work done} &= \text{Change in K.E} &= \Delta \text{ K.E} \\ &= \text{Final K.E} - \text{Initial K.E} \\ &= \frac{1}{2}mv^2 - \frac{1}{2}mu^2\end{aligned}$$

$$\begin{aligned}\Rightarrow F \cdot S &= \frac{1}{2}mv^2 - \frac{1}{2}mu^2 & [W = F \cdot S] \\ \Rightarrow m \cdot a \cdot \frac{(v^2 - u^2)}{2a} &= \frac{1}{2}m(v^2 - u^2) &= \left(\because S = \frac{(v^2 - u^2)}{2a} \right)\end{aligned}$$

$$\Rightarrow v^2 - u^2 = v^2 - u^2$$

Hence, the proved.

POWER

- The rate of work done is known as Power.

OR

The rate at which work is done by Force is called Power delivered by the Force.

OR

The rate which the energy is transferred by an object (like a machine) is called power delivered by that object.

i.e, $\text{POWER} = \frac{\text{Work Done}}{\text{Time}}$

→ Power is denoted by 'P'.

$$\therefore P = \frac{W}{t}$$

$$\Rightarrow W = p \cdot t$$

$$\Rightarrow t = \frac{W}{p}$$

→ The S.I unit of Power $= \frac{\text{S.I unit of Work}}{\text{S.I unit of Time}}$
 $= \text{Joule / Second}$

→ The S.I unit of Power J/s.

Note:

➤ 1 J/s = 1 Watt

$\Rightarrow 1 \text{ J/s} = 1 \text{ W}$

Note:

- “Watt” named in honored of “James Watt”.

▪ **The other unit of power**

- a) 1 Kilowatt (Kw) = 1000 W
- b) 1 Megawatt (M W) = 10^6 W
- c) 1 Horsepower (1 HP) = 746 W

Note:

- The commercial unit of power is 1 kWh / 1 unit / 1 B.O.T Unit.

[B.O.T Unit] → Board of trade Unit

• **1 Unit / 1 B.O.T Unit is defined as**

- If an electrical device of 1000 W (1 kW) is used to an hour then electrical energy consumption is 1 Kwh.

$$\begin{aligned}\text{i.e, } 1 \text{ kWh} &= 1 \text{ kwh} \\ &= 1000 \text{ w} \times 1 \text{ h} \\ &= 1000 \text{ J/s} \times 3600 \text{ s} \\ &= 3600000 \text{ J} \\ &= 3.6 \times 10^6 \text{ J}\end{aligned}$$

$$\therefore 1 \text{ Unit} = 3.6 \times 10^6 \text{ J}$$

- Q. In an office, four fans of 100 w and 5 bulbs of 60 w are used every day for 8 hours. Find the amount of electrical energy bill in one month. If charge of one unit is Rs. 7.

$$\begin{aligned}\text{Ans Total power used in a day} &= (4 \times 100 + 5 \times 60) \times 8 \\ &= (400 + 300) \times 8 \\ &= 700 \times 8 \\ &= 5600 \text{ w}\end{aligned}$$

$$\begin{aligned}\therefore \text{Total power used in 30 days} &= 30 \times 5600 \\ &= 168000 \text{ watt} \\ &= \frac{168000}{1000} = 168 \text{ units}\end{aligned}$$

$$\therefore \text{Price of 1 unit} = 7 \text{ Rs.}$$

$$\begin{aligned}\therefore \text{Price of 168 units} &= 168 \times 7 \\ &= 1176 \text{ Rs.}\end{aligned}$$